

Question number	Answer	Additional guidance	Mark
1(a)	An explanation that includes the following points: • (magnesium ions are) components of chlorophyll (1) • which <u>absorbs</u> {light / photons} (1)	Examiners will need to check other roles if seen	(2)

- component of (leaf) ribosomes
- necessary for translation

- cofactor / allosteric modulator
- enabling {carboxylases / phosphatases / protein kinases / RNA polymerase / ATP synthases} to function

One of these would give 1 mark only:

involved in {phloem loading / sucrose transport in phloem / improved root : shoot ratio / vigorous root growth}
 supports {transpiration rate / stomatal conductance / carbon dioxide fixation / non-cyclic photophosphorylation}
 stabilises DNA
 involved in ATP synthesis

Question number	Answer	Additional guidance	Mark
1(b)	An explanation that includes four of the following points: - uptake of magnesium ions levels off without oxygen but continues to increase with oxygen (1) - uptake of magnesium ions increases as the ions can be taken up against the concentration gradient (1) {ATP / energy} is needed for {this / active transport} (1) - and oxygen is used in aerobic respiration to produce this ATP (1) - if {active transport was not involved / uptake was passive} the rate would have been the same in both groups (1)	ACCEPT more magnesium ions are taken up in presence of oxygen rate of uptake is faster in presence of oxygen NB piece together PIECE TOGETHER	(4)